

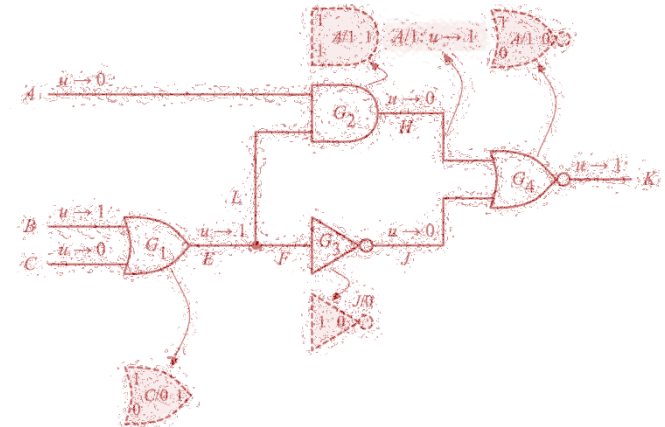
Fault Simulation

- Introduction
- Fault simulation techniques
 - ◆ Serial fault simulation
 - ◆ Parallel fault simulation (1965)
 - ◆ PPSFP (1985)
 - ◆ Deductive fault simulation (1972)
 - ◆ Concurrent fault simulation (1974)
 - ◆ Differential fault simulation (1989)

- * Concept

- * Example

- Alternatives to fault simulation
- Issues of fault simulation
- Concluding remarks



What is Fault Simulation?

- Given m faults, n patterns
 - simulate $m \times n$ **2D matrix**
- G_i = Good circuit with **test pattern V_i** applied
- $B_{k,i}$ = Bad circuit of **fault k** , with **test pattern V_i** applied

	V_1	...	V_i	V_{i+1}	...	V_n
Good	G_1	...	G_i	G_{i+1}	...	G_n
Bad ₁	$B_{1,1}$	...	$B_{1,i}$	$B_{1,i+1}$	...	$B_{1,n}$
Bad ₂	$B_{2,1}$	...	$B_{2,i}$	$B_{2,i+1}$	...	$B_{2,n}$
...	...	...	...	...	...	...
Bad _k	$B_{k,1}$	...	$B_{k,i}$	$B_{k,i+1}$	...	$B_{k,n}$
Bad _{k+1}	$B_{k+1,1}$	...	$B_{k+1,i}$	$B_{k+1,i+1}$	...	$B_{k+1,n}$
...	...	...	...	...	...	...
Bad _m	$B_{m,1}$	...	$B_{m,i}$	$B_{m,i+1}$	...	$B_{m,n}$

Previous Problem (1) PPSFP

- PPSFP simulate bad circuits $B_{k,x}$ together
- Advantage: parallel simulation is fast

	V_1	...	V_i	V_{i+1}	...	V_n
Good	G_1	...	G_i	G_{i+1}	...	G_n
Bad ₁	$B_{1,1}$	...	$B_{1,i}$	$B_{1,i+1}$	...	$B_{1,n}$
Bad ₂	$B_{2,1}$	...	$B_{2,i}$	$B_{2,i+1}$	...	$B_{2,n}$
...	...	...	...	...	...	...
Bad _k	$B_{k,1}$	...	$B_{k,i}$	$B_{k,i+1}$	...	$B_{k,n}$
Bad _{k+1}	$B_{k+1,1}$	...	$B_{k+1,i}$	$B_{k+1,i+1}$	...	$B_{k+1,n}$
...	...	...	...	...	...	...
Bad _m	$B_{m,1}$	...	$B_{m,i}$	$B_{m,i+1}$	...	$B_{m,n}$

Previous Problem (1) PPSFP

- However, each simulation **restarts** from good circuit G_x state
- Problem : **sequential circuit states not preserved**
 - ♦ **NOT** applicable to sequential circuits

	V_1	...	V_i	V_{i+1}	...	V_n
Good	G_1	...	G_i	G_{i+1}	...	G_n
Bad ₁	$B_{1,1}$	...	$B_{1,i}$	$B_{1,i+1}$	...	$B_{1,n}$
Bad ₂	$B_{2,1}$	...	$B_{2,i}$	$B_{2,i+1}$	...	$B_{2,n}$
...	...	...	...	...	...	...
Bad _k	$B_{k,1}$	...	$B_{k,i}$	$B_{k,i+1}$	...	$B_{k,n}$
Bad _{k+1}	$B_{k+1,1}$	...	$B_{k+1,i}$	$B_{k+1,i+1}$	...	$B_{k+1,n}$
...	...	...	...	...	...	...
Bad _m	$B_{m,1}$	...	$B_{m,i}$	$B_{m,i+1}$	...	$B_{m,n}$

Previous Problem (2) Concurrent

- Concurrent simulate bad circuits $B_{x,i+1}$ from previous $B_{x,i}$
- Advantage: preserve circuit states
 - ♦ applicable to sequential circuits

	V_1	...	V_i	V_{i+1}	...	V_n
Good	G_1	...	G_i	G_{i+1}	...	G_n
Bad ₁	$B_{1,1}$	...	$B_{1,i}$	$B_{1,i+1}$	...	$B_{1,n}$
Bad ₂	$B_{2,1}$	...	$B_{2,i}$	$B_{2,i+1}$	...	$B_{2,n}$
...	...	...	...	...	...	...
Bad _k	$B_{k,1}$	...	$B_{k,i}$	$B_{k,i+1}$	...	$B_{k,n}$
Bad _{k+1}	$B_{k+1,1}$	...	$B_{k+1,i}$	$B_{k+1,i+1}$	...	$B_{k+1,n}$
...	...	...	...	...	...	...
Bad _m	$B_{m,1}$	...	$B_{m,i}$	$B_{m,i+1}$	...	$B_{m,n}$

Previous Problem (2) Concurrent

- Need to store **ALL** states of $B_{k,i}$ so that we can simulate $B_{k,i+1}$
- Problem: **memory management difficult**



	V_1	...	V_i	V_{i+1}	...	V_n
Good	G_1	...	G_i	G_{i+1}	...	G_n
Bad ₁	$B_{1,1}$	...	$B_{1,i}$	$B_{1,i+1}$	...	$B_{1,n}$
Bad ₂	$B_{2,1}$	...	$B_{2,i}$	$B_{2,i+1}$	...	$B_{2,n}$
...	...	...	...	...	...	...
Bad _k	$B_{k,1}$	...	$B_{k,i}$	$B_{k,i+1}$	...	$B_{k,n}$
Bad _{k+1}	$B_{k+1,1}$	...	$B_{k+1,i}$	$B_{k+1,i+1}$	...	$B_{k+1,n}$
...	...	...	...	...	...	...
Bad _m	$B_{m,1}$	...	$B_{m,i}$	$B_{m,i+1}$	...	$B_{m,n}$

Quiz

Q: Which of the following is CORRECT?

- A) PPSFP is applicable to sequential circuits**
- B) concurrent fault simulation is applicable to sequential circuits**
- C) PPSFP wastes memory to store FF states**

Differential Fault Sim. [Cheng 89]

- **Differential fault simulation (DSIM)**
 - ♦ Simulate differences from the last simulated circuit good or bad circuit
- Better than PPSFP
 - ♦ Simulate one pattern at a time
 - * Preserve sequential circuit states
- Better than concurrent
 - ♦ Record circuit states DIFFERENCE between $B_{k,i}$ and $B_{k+1,i}$
 - * Reduce memory
- Up to 12 times faster than concurrent, PPSFP

DSIM Combines Concurrent and PPSFP

DSIM Idea

- Simulates one pattern at a time
- First good circuit, then faulty circuits

	V_1	...	V_i	V_{i+1}	...	V_n
Good	G_1	...	G_i	G_{i+1}	...	G_n
Bad ₁	$B_{1,1}$	...	$B_{1,i}$	$B_{1,i+1}$	...	$B_{1,n}$
Bad ₂	$B_{2,1}$	...	$B_{2,i}$	$B_{2,i+1}$	...	$B_{2,n}$
...	...	...	...	...	...	...
Bad _k	$B_{k,1}$	...	$B_{k,i}$	$B_{k,i+1}$	...	$B_{k,n}$
Bad _{k+1}	$B_{k+1,1}$	...	$B_{k+1,i}$	$B_{k+1,i+1}$	...	$B_{k+1,n}$
...	...	...	...	...	...	...
Bad _m	$B_{m,1}$	...	$B_{m,i}$	$B_{m,i+1}$	...	$B_{m,n}$

DSIM Idea (2)

- Simulates one pattern at a time
- First good circuit, then faulty circuits

	V_1	...	V_i	V_{i+1}	...	V_n
Good	G_1	...	G_i	G_{i+1}	...	G_n
Bad ₁	$B_{1,1}$	...	$B_{1,i}$	$B_{1,i+1}$	...	$B_{1,n}$
Bad ₂	$B_{2,1}$	...	$B_{2,i}$	$B_{2,i+1}$	...	$B_{2,n}$
...	...	...	...	...	...	...
Bad _k	$B_{k,1}$	...	$B_{k,i}$	$B_{k,i+1}$	...	$B_{k,n}$
Bad _{k+1}	$B_{k+1,1}$	...	$B_{k+1,i}$	$B_{k+1,i+1}$	...	$B_{k+1,n}$
...	...	...	...	...	...	...
Bad _m	$B_{m,1}$	...	$B_{m,i}$	$B_{m,i+1}$	...	$B_{m,n}$

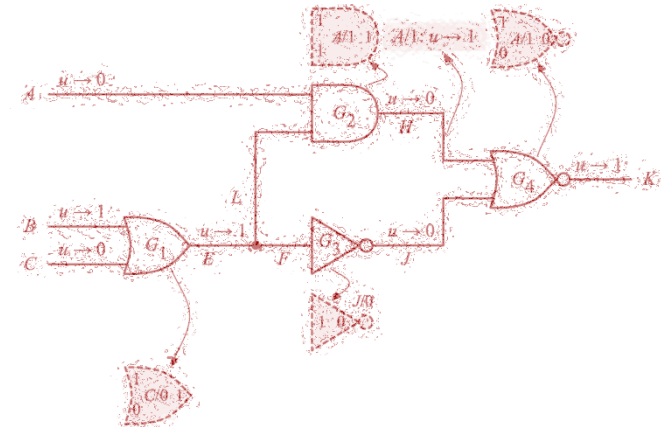
DSIM Idea (3)

- Recover sequential circuit state
 - from last simulated circuit

	V_1	...	V_i	V_{i+1}	...	V_n
Good	G_1	...	G_i	G_{i+1}	...	G_n
Bad ₁	$B_{1,1}$	...	$B_{1,i}$	$B_{1,i+1}$	...	$B_{1,n}$
Bad ₂	$B_{2,1}$	...	$B_{2,i}$	$B_{2,i+1}$	...	$B_{2,n}$
...	...	...	...	...	...	...
Bad _k	$B_{k,1}$	...	$B_{k,i}$	$B_{k,i+1}$	...	$B_{k,n}$
Bad _{k+1}	$B_{k+1,1}$	...	$B_{k+1,i}$	$B_{k+1,i+1}$	...	$B_{k+1,n}$
...	...	...	...	...	...	...
Bad _m	$B_{m,1}$	...	$B_{m,i}$	$B_{m,i+1}$	...	$B_{m,n}$

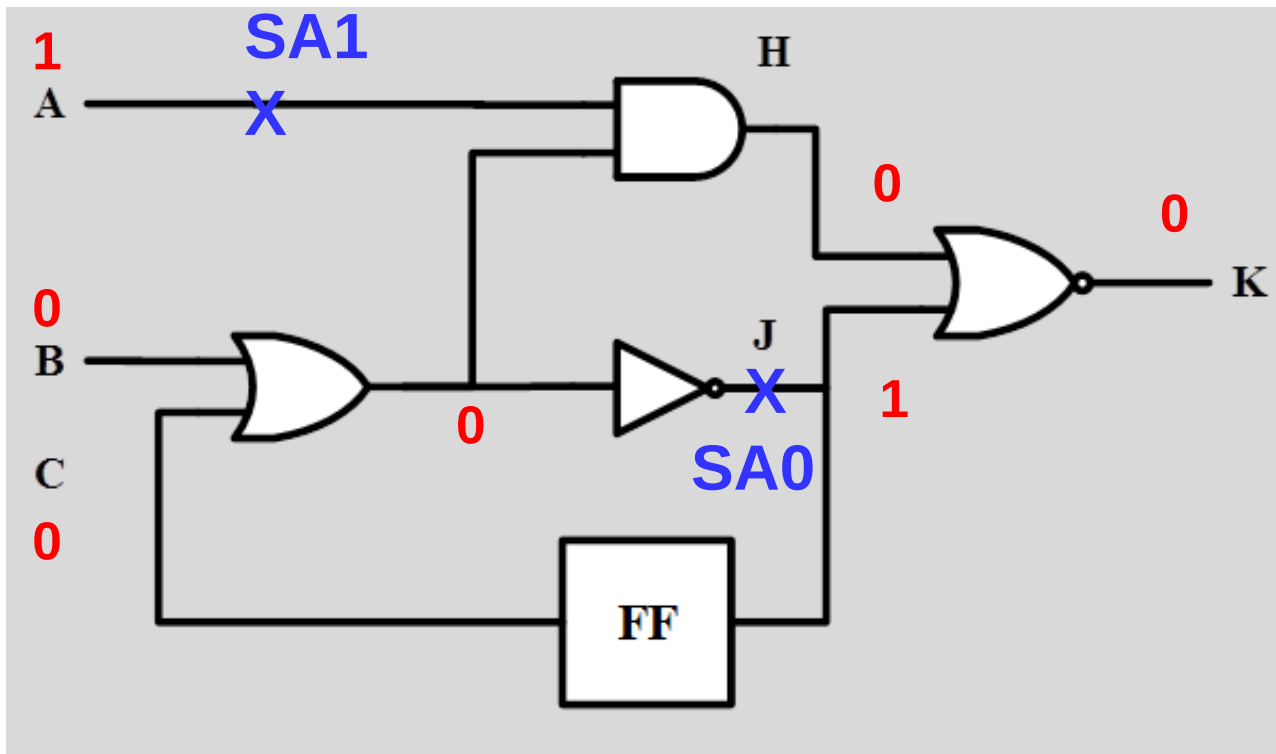
Fault Simulation

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 - ◆ Concurrent fault simulation (1974)
 - ◆ Differential fault simulation (1989)
 - * **Concept**
 - * **Example**
- Alternatives to fault simulation
- Issues of fault simulation
- Concluding remarks



Example

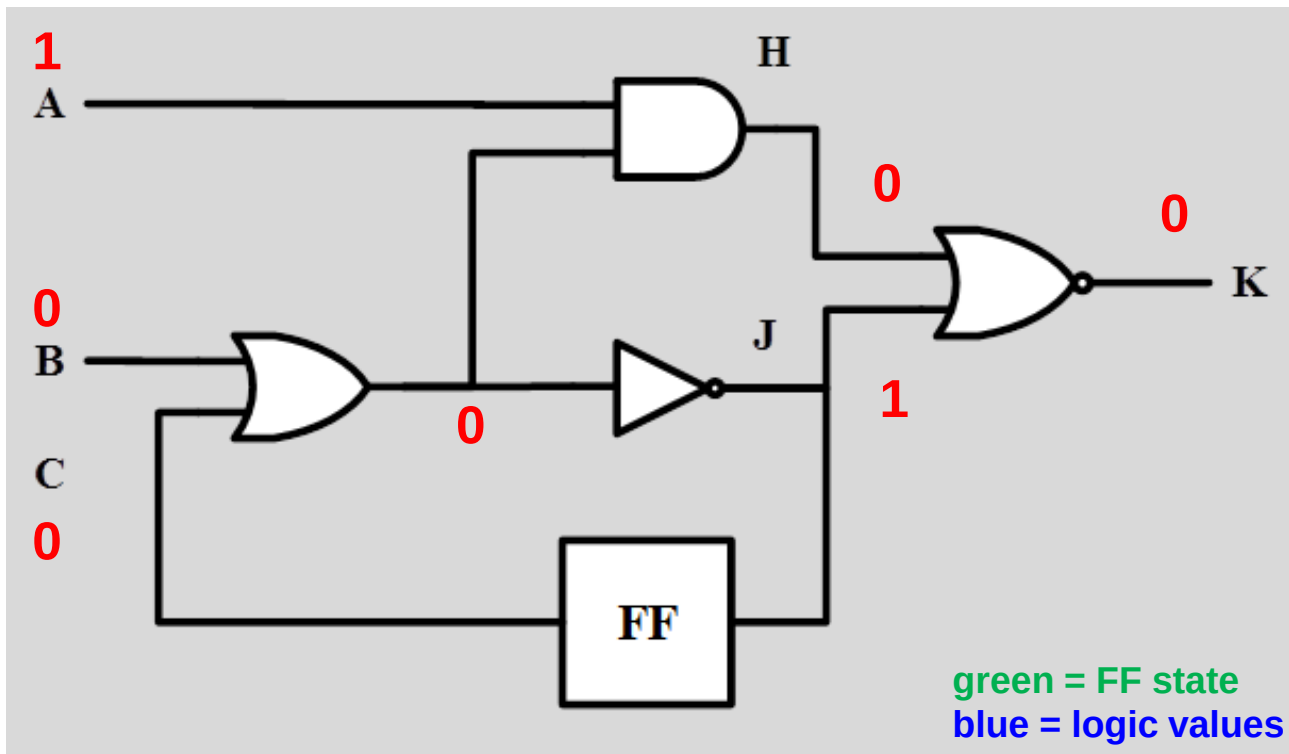
- Apply two patterns: AB=10 , 00 . Initially FF state =0
- Consider two faults: A SA1, J SA0
 - ♦ No fault dropping for demo purpose



V_1 AB=10	V_2 AB=00
G_1 FF=0 (given)	G_2
$B_{1,1}$ A SA1	$B_{1,2}$ A SA1
$B_{2,1}$ J SA0	$B_{2,2}$ J SA0

Example (1)

- Current state $FF=0$. Apply $AB=10$ to good circuit
 - $FF^+=1 \rightarrow$ store next FF state
 - $K=0 \rightarrow$ store current PO

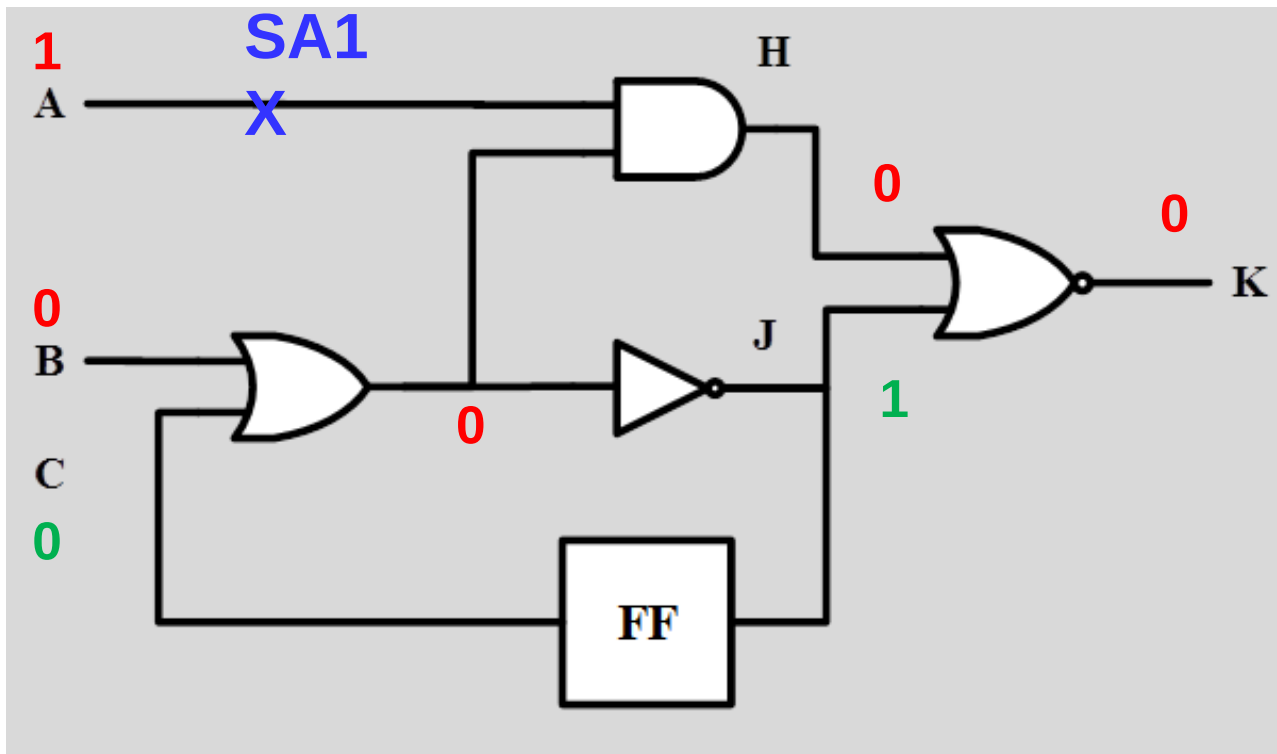


V_1 $AB=10$	V_2 $AB=00$
G_1 $K=0$ $FF^+=1$	G_2
$B_{1,1}$	$B_{1,2}$
$B_{2,1}$	$B_{2,2}$

$FF^+ = \text{next FF}$

Example (2)

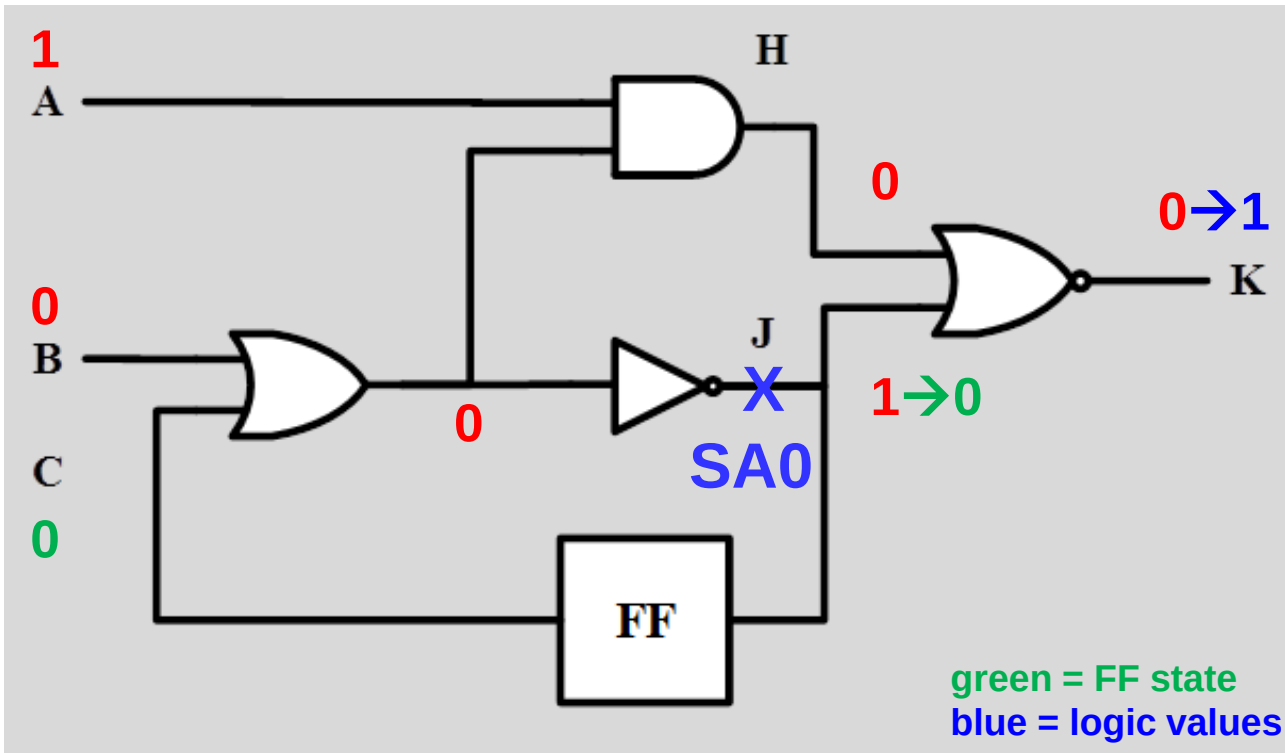
- Inject A SA1, C=0
 - ♦ $FF^+=1$ same as $G_1 \rightarrow$ no need to store state difference
 - ♦ $K=0$ same as $G_1 \rightarrow$ fault undetected



V_1 $AB=10$	V_2 $AB=00$
G_1 $K=0$ $FF^+=1$	G_2
$B_{1,1}$	$B_{1,2}$
$B_{2,1}$	$B_{2,2}$

Example (3)

- Remove A SA1. Inject J SA0
 - $FF^+=1$ different from $B_{1,1} \rightarrow$ store state difference from $B_{1,1}$
 - K different from $G_1 \rightarrow$ fault detected



V_1 $AB=10$	V_2 $AB=00$
G_1 $K=0$ $FF^+=1$	G_2
$B_{1,1}$	$B_{1,2}$
$B_{2,1}$ $K=1$ $FF^+=0$	$B_{2,2}$

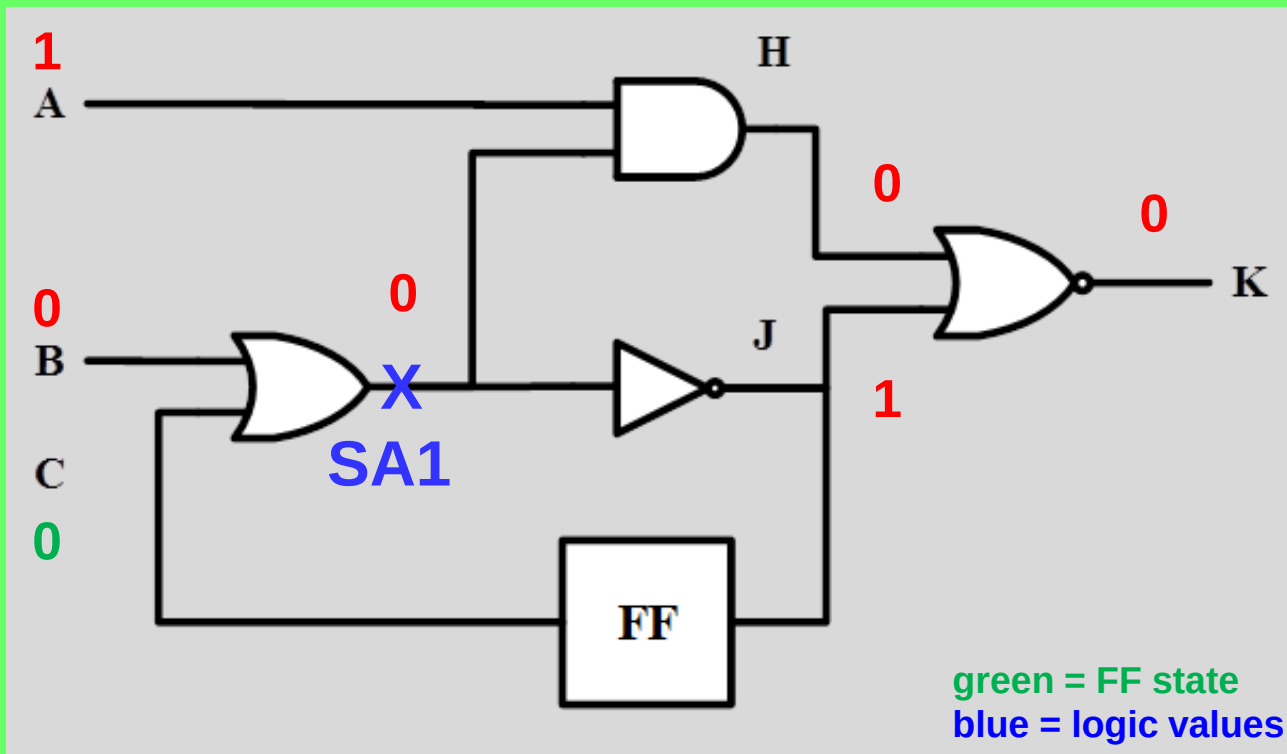
$FF^+ = \text{next FF}$

QUIZ

Q1: Consider a third fault, please fill in ?

Q2: What is K? Is the fault detected?

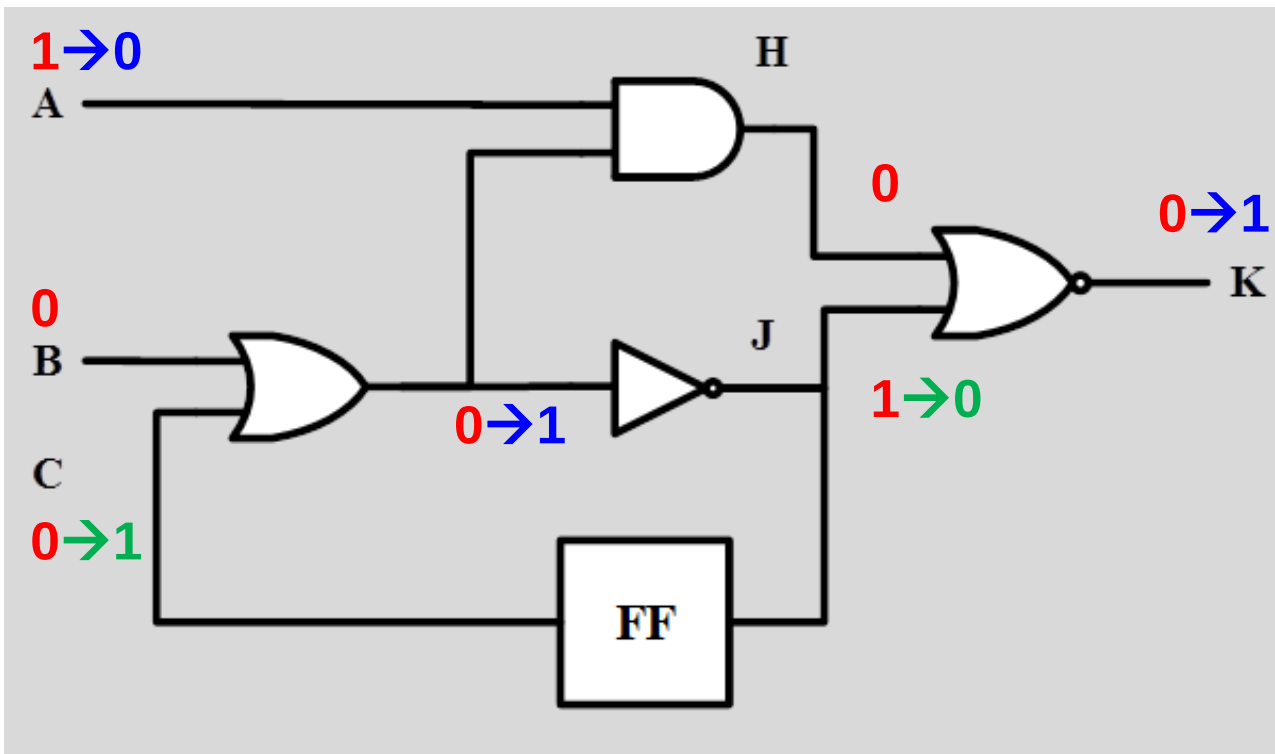
Q3: What is FF^+ ? Will this be recorded?



V_1 $AB=10$	V_2 $AB=00$
G_1 $K=0$ $FF^+=1$	G_2
$B_{1,1}$	$B_{1,2}$
$B_{2,1}$ $K=1$ $FF^+=0$	$B_{2,2}$
$B_{3,1}$ $K=?$ $FF^+=?$	

Example (4)

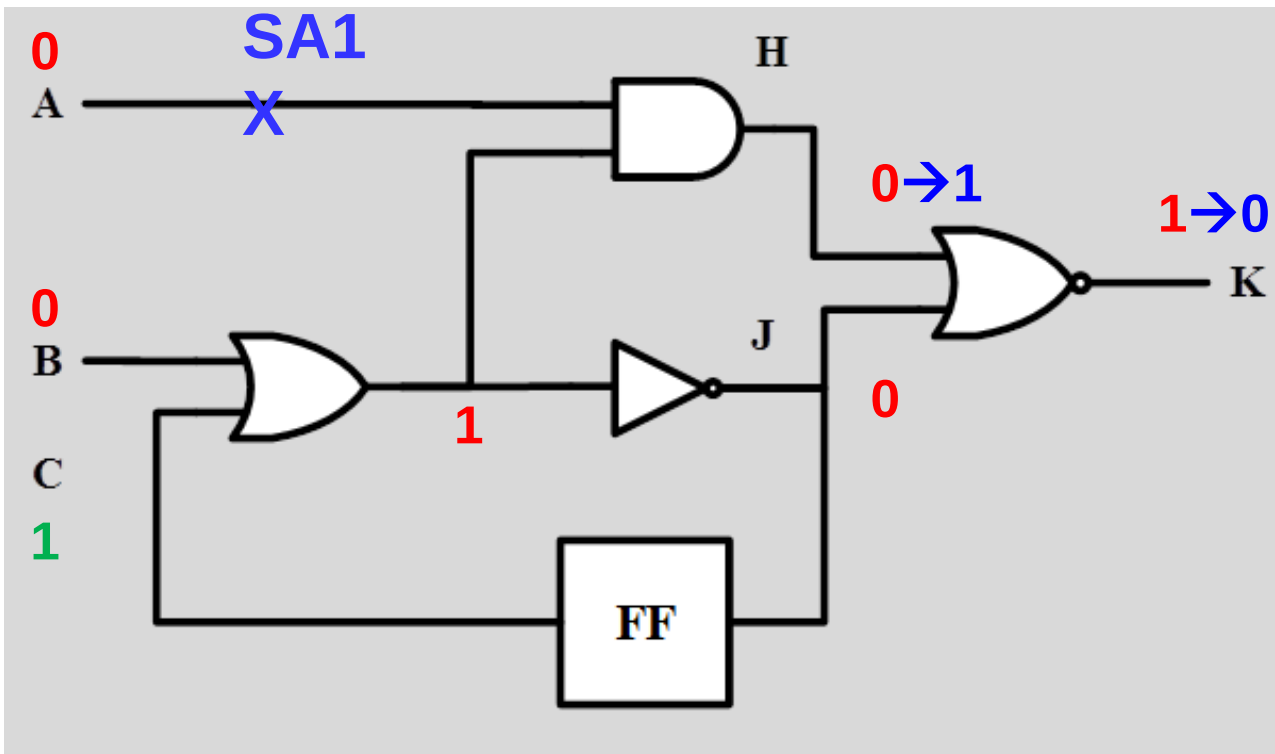
- Remove J SA0. Apply V_2 : $A=1 \rightarrow 0$. Restore good state: $C=0 \rightarrow 1$
 - $FF^+=0 \rightarrow$ store next FF state
 - $K=1 \rightarrow$ store current PO



V_1 $AB=10$	V_2 $AB=00$
G_1 $K=0$ $FF^+=1$	G_2 $K=1$ $FF^+=0$
$B_{1,1}$	$B_{1,2}$
$B_{2,1}$ $FF^+=0$	$B_{2,2}$

Example (5)

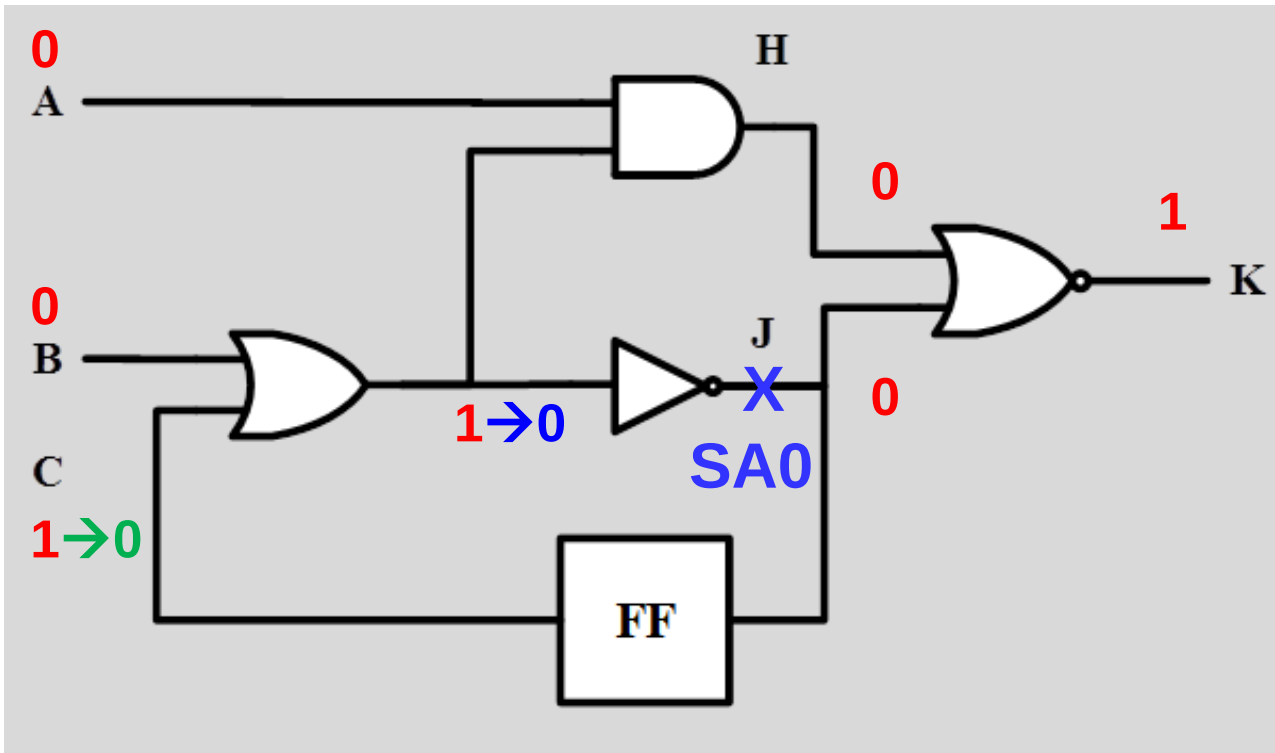
- Inject A SA1. A=0, C=1 (same as good)
 - FF⁺=0 same as G₂ → no need to store FF state difference
 - K different from G₂ → fault detected



V ₁ AB=10	V ₂ AB=00
G ₁	G ₂ K=1 FF ⁺ =0
B _{1,1}	B _{1,2} K=0
B _{2,1} FF ⁺ =0	B _{2,2}

Example (6)

- Restore state from $B_{2,1}$: $C=1 \rightarrow 0$
 - ♦ $FF^+=0$ same as $B_{1,2} \rightarrow$ no need to store FF state difference
 - ♦ K same as $G_2 \rightarrow$ fault undetected



V_1 $AB=10$	V_2 $AB=00$
G_1	G_2 $K=1$ $FF^+=0$
$B_{1,1}$	$B_{1,2}$
$B_{2,1}$ $FF^+=0$	$B_{2,2}$

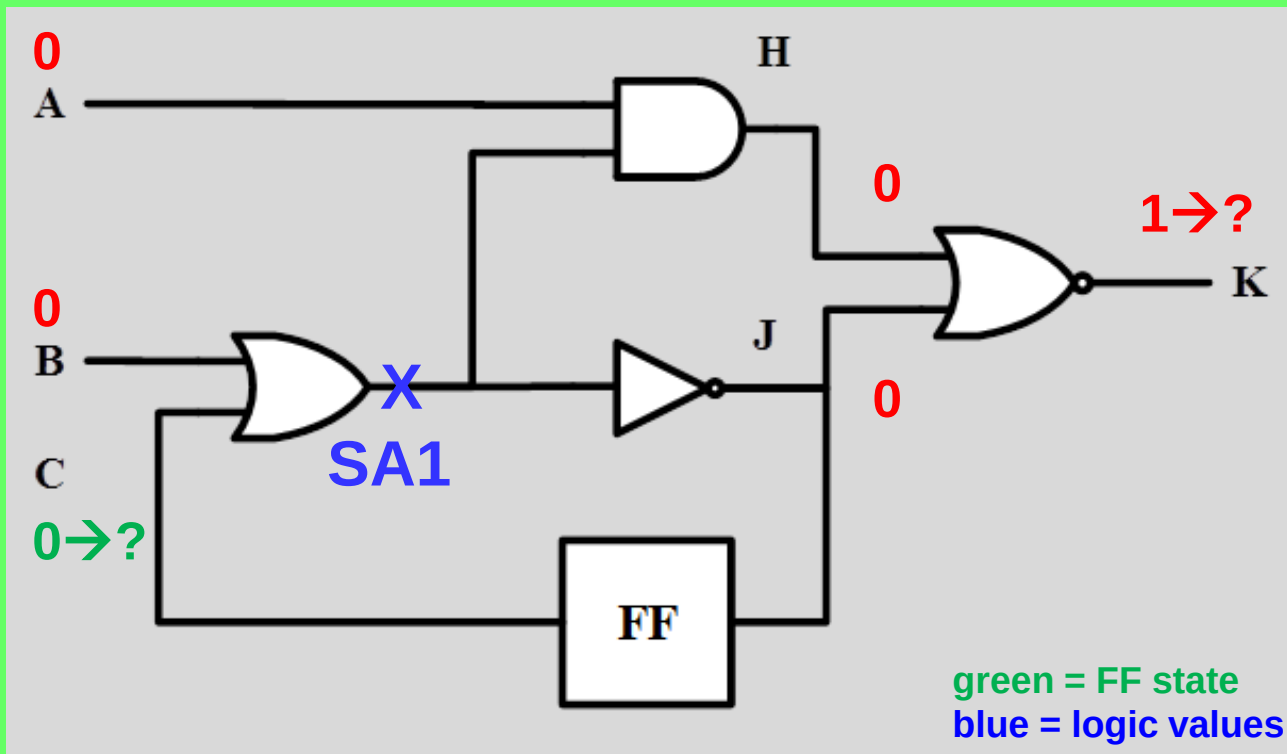
Very Small Memory Required

QUIZ

Q1: Consider a third fault, please fill in ?

Q2: What is K? Is the fault detected?

Q3: What is FF^+ ? Will this be recorded?



V_1 $AB=10$	V_2 $AB=00$
G_1	G_2 $K=1$ $FF^+=0$
$B_{1,1}$	$B_{1,2}$
$B_{2,1}$ $FF^+=0$	$B_{2,2}$
$B_{3,1}$	$B_{3,2}$ $K=?$ $FF^+=?$

DFSIM Algorithm

```

foreach test vector  $V_i$ 
  if ( $V_i$  is first vector)
    initialize ckt states
  else
    remove previously injected fault
    recover current states
    set  $V_i$  pattern at primary inputs
    event-driven simulation
    store PO
    sensitized output counter =0
    foreach undetected faulty machine  $B_j$ 
      remove previous injected fault
      recover current states
      inject current fault
      event-driven simulation
      if sensitized output counter >0
        drop the fault
  
```

Restore a circuit state like PPSFP.

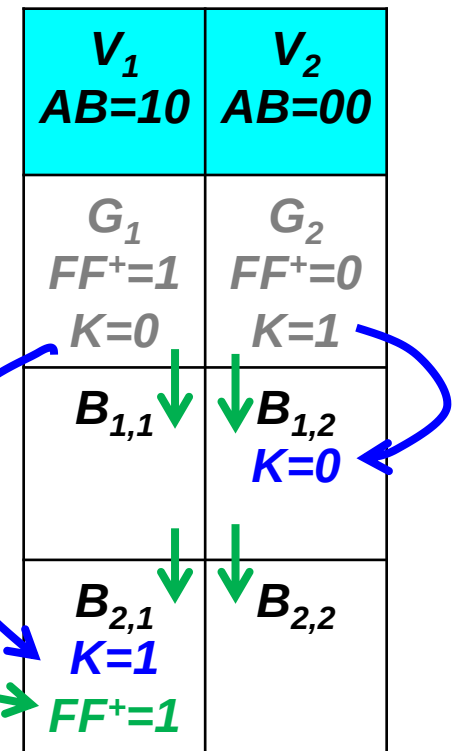
FF+ serves as a primary input to a gate in the next vector.

V_1 $AB=10$	V_2 $AB=00$
G_1 $FF^+=1$ $K=0$	G_2 $FF^+=0$ $K=1$
$B_{1,1}$	$B_{1,2}$
$B_{2,1}$	$B_{2,2}$

DFSIM Algorithm

```

foreach test vector  $V_i$ 
  if ( $V_i$  is first vector)
    initialize ckt status
  else
    remove previously injected fault
    recover current states
    set  $V_i$  pattern at primary inputs
    event-driven simulation
    store PO
    sensitized output counter =0
    foreach undetected faulty machine  $B_j$ 
      remove previous injected fault
      recover current states
      inject current fault
      event-driven simulation
      if sensitized output counter >0
        drop the fault
  
```



Summary

- Differential fault simulation
 - ◆ Simulate one vector, one circuit at a time
 - ◆ Restore **state difference** from previous circuit simulated
- Advantages
 - 😊 Applicable to **sequential circuits**
 - * Better than PPSFP fault simulation
 - 😊 Require very **small memory**
 - * Better than concurrent fault simulation
 - 😊 **Event-driven** simulation
 - * Handles delay faults

DSIM Very Useful for Seq. Ckt.

FFT

- Q: If a fault is dropped, how can we restore FF state of next fault?
 - ♦ why do not we record FF state of all Good FF

	V_1	...	V_i	V_{i+1}	...	V_n
Good	G_1	...	G_i	G_{i+1}	...	G_n
Bad ₁	$B_{1,1}$	...	$B_{1,i}$	$B_{1,i+1}$	...	$B_{1,n}$
Bad ₂	$B_{2,1}$	...	$B_{2,i}$	$B_{2,i+1}$	...	$B_{2,n}$
...	...	...	...	...	...	...
Bad _k	$B_{k,1}$	...	$B_{k,i}$	$B_{k,i+1}$	...	$B_{k,n}$
Bad _{k+1}	$B_{k+1,1}$	...	$B_{k+1,i}$	$B_{k+1,i+1}$	...	$B_{k+1,n}$
...	...	...	...	...	...	...
Bad _m	$B_{m,1}$	...	$B_{m,i}$	$B_{m,i+1}$	...	$B_{m,n}$

